

Model-Free Predictive Current Control Using Extended Affine Ultralocal for PMSM Drives

Yao Wei , *Member, IEEE*, Hector Young , *Senior Member, IEEE*,
Dongliang Ke , *Senior Member, IEEE*, Fengxiang Wang , *Senior Member, IEEE*,
and José Rodríguez , *Life Fellow, IEEE*

Abstract—Ultralocal is always used to solve the problem of weak robustness in predictive control to resist the influences caused by some time-varying physical parameters in the motor driving system. However, the achieved model cannot reflect the motion characteristics of the motor accurately and timely. Considering the influences on the model accuracy caused by nonlinear terms, a model-free predictive current control (MF-PCC) using extended affine ultralocal is proposed in this article for solving these issues. A data-driven model of extended affine ultralocal is built containing a two-order term based on affine arithmetic, and all coefficients in the model are online estimated by the recursive least square algorithm, including the input gain in the MF-PCC strategy based on the conventional ultralocal. The proposed method is completely independent from prior knowledge of the physical parameters of the controlled system. The proposed method is applied to a permanent magnet synchronous motor driving system, and the simulation and experimental results demonstrate the effectiveness and advantages including improved current quality and robustness compared with the MF-PCC strategy based on the conventional ultralocal model.

Index Terms—Affine arithmetic, extended affine ultralocal model, model-free predictive control, recursive least square (RLS) algorithm.

I. INTRODUCTION

WITH the development of the industry, the applications of the motor driving system become wider and the

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Yao Wei, Dongliang Ke, and Fengxiang Wang are with the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Quanzhou 362200, China (e-mail: yao.wei@fjirms.ac.cn; dongliang.ke@fjirms.ac.cn; fengxiang.wang@fjirms.ac.cn).

Hector Young is with the Department of Electrical Engineering, Universidad de La Frontera, Temuco 481230, Chile (e-mail: hector.young@ufrontera.cl).

José Rodríguez is with the Faculty of Engineering, Universidad San Sebastian, Santiago 8370146, Chile (e-mail: jose.rodriguezp@uss.cl).

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requirements for the control strategy become more strict. Since conventional control strategies cannot satisfy the strict requirements fully, model predictive control (MPC) is highly regarded in the motor driving system due to its great dynamics, easy understanding, and multiple optimizations [1]. To realize different primary objectives of the system, MPC can be mainly divided into predictive current control (PCC), predictive torque control, and predictive speed control [2], [3]. MPC includes two parts, including the discrete-time model adopted to predict the variables and the cost function used to minimize errors among the objectives [4]. Since the predicted variables and achieved performances are strongly related to a priori model including time-varying physical parameters of the motor, and the system continuously operates under the condition of parameter mismatches, the weak robustness of the typical MPC strategy limits its popularization [5].

To improve this issue, the influences caused by the parameter mismatches are seen as a disturbance, and online estimated and compensated to the control variables by some observers, such as disturbance observer [6], sliding mode observer (SMO) [7], and extended state observer (ESO) [8]. Some advanced observers are presented and used to estimate disturbances to improve the observation accuracy. Considering the closed-loop predictive model, an observer-predictor-based predictive control is presented to accurately predict the variables [9]. A modified adaptive full-order observer is designed to improve the operating state within the low-speed region of the motor [10]. Besides, parameter identification is another effective method to solve this issue, especially for the full model or parameter identification methods, all possible time-varying parameters are estimated online to reduce the adverse impacts [11], [12]. Although these methods can enhance the robustness effectively, the influences of parameter mismatches cannot be fully eliminated due to the need for a priori model.

Since there might be no access to confirm all parameters of the plant in time, model-free predictive control makes the plant fit as an online updating data-driven model to remove the dependence on physical parameters, as well as their influences, and to guarantee the performance of the system with unmodeled dynamics [13], [14]. Due to its easy implementation and simple structure, the ultralocal model is commonly used in the motor driving system, in which all unknown parts of the motor are summarized as a single variable and estimated by an algebraic algorithm [15]. To further simplify the calculation process and

TABLE II
PRIMARY SIMULATION PARAMETERS

Symbols	Quantity	Values
R_s	Stator resistance	0.637 Ω
L_{sd}	Stator inductance on d -axis	10 mH
L_{sq}	Stator inductance on q -axis	14.5 mH
p	Number of pole pairs	4
ψ_m	Magnetic flux linkage	0.31 Wb
J	Rotor inertia	0.005 kg.m ²
V_{dc}	dc voltage	540 V
T_s	Sampling period	62.5 μ s
μ	Weighting factor	2
λ	Forgetting factor	1
n	length of errors	3
T_{rated}	Rated torque	23 N.m
n_{rated}	Rated speed	2000 r/min

follows:

$$i_{sd}(k+1) = \left(1 - \frac{R_s T_s}{L_{sd}}\right) i_{sd}(k) + \frac{L_{sq}}{L_{sd}} T_s \omega_r i_{sq}(k) + \frac{T_s}{L_{sd}} v_{sd} \quad (4)$$

$$i_{sq}(k+1) = \left(1 - \frac{R_s T_s}{L_{sq}}\right) i_{sq}(k) - T_s \omega_r i_{sd}(k) - \frac{L_{sd}}{L_{sq}} \psi_m \omega_r i_{sd}(k) + \frac{T_s}{L_{sq}} v_{sq} \quad (5)$$

where $i_{sd}(k)$ and $i_{sq}(k)$ are the discrete-time variables at the K th sampling period.

B. Control Strategy

The primary objective of the PCC strategy is to make the current components follow their respective references to obtain the minimum error between the components and references. To realize this objective, the cost function is designed as follows:

$$J = (i_{sd}(k+1))^2 + \mu (i_{sq}^* - i_{sq}(k+1))^2 \quad (6)$$

where i_{sq}^* is the reference of the q -axis current component, and μ is the weighting factor reflecting the importance of objectives, which is tuned by the branch and bound algorithm. The first term of the cost function makes the d -axis current component operate around its reference, in which it is assumed that the reference for the d -axis current is zero. The second term has the same function for the q -axis component.

All candidate vectors are substituted into the model and cost function in turn, to find the optimal vector having the minimum cost function value. The switching state of the optimal vector affects the drive to adjust its operating state.

Since a physical model is required in the conventional PCC, it is named model-based PCC strategy reflecting this strategy includes physical parameters and structure of the plant.

III. EXTENDED AFFINE ULTRALOCAL-BASED MF-PCC

A. Basic Ultralocal Structure

The basic one-order ultralocal structure of the motor is as follows:

$$\frac{di_s}{dt} = \alpha u_s + F \quad (7)$$

where α is the input gain, which can be selected as $1/L_{sq}$, $i_s = [i_{sd}, i_{sq}]^T$, and $u_s = [u_{sd}, u_{sq}]^T$. All influences caused by the parameter mismatches are summarized as a single variable as $F = [F_d, F_q]^T$, which needs to be estimated online.

The following LESO is seen as an example to estimate $\hat{F}$ online [16]:

$$\begin{cases} e = z_1(k) - i_s \\ z_1(k+1) = z_1(k) + T_s(z_2(k) + \alpha u_s - \beta_1 e) \\ z_2(k+1) = z_2(k) - T_s \beta_2 e \end{cases} \quad (8)$$

where z_1 is $\hat{i}_s$, z_2 is $\hat{F}$, and β_1 and β_2 are the feedback gains tuned according to the required bandwidth.

The phase current total harmonic distortions (THDs) and speed integrated time and absolute errors (ITAEs) with different fixed input gains α are shown in Fig. 1. It shows that the optimal α with the minimum THD and ITAE values is continuously changed under different operating conditions. An online selection method for α is necessary to achieve a better current quality.

B. Extended Affine Ultralocal Structure

1) Two-Order Extended Affine Ultralocal Structure: The estimated variable $\hat{F}$ can be seen as an affine operator, and the model is improved as the following formula considering the two-order sensitivity function:

$$\frac{d\hat{e}}{dt} = \alpha u_s + F_0 + F_1 \hat{e} + \hat{e}^T F_2 \hat{e} \quad (9)$$

where F_0 is the expected operating point, and F_1 and F_2 are the sensitive coefficients reflecting the linear and nonlinear relationships between the affine operator $\hat{F}$ and affine variable $\hat{e}$. These matrices define as follows:

$$\begin{cases} F_0 = \text{diag}(F_{0d}, F_{0q}) \\ F_1 = \text{diag}(F_{1d}, F_{1q}) \\ F_2 = \text{diag}(F_{2d}, F_{2q}) \end{cases} \quad (10)$$

and the affine variable $\hat{e}$ is designed as a group of the normalized current errors as follows:

$$\hat{e} = \left[\frac{\xi_d - \min_n(\xi_d)}{\max_n(\xi_d) - \min_n(\xi_d)}, \frac{\xi_q - \min_n(\xi_q)}{\max_n(\xi_q) - \min_n(\xi_q)} \right]^T \in [-1, 1] \quad (11)$$

where the subscript n is the length of the errors, which is selected as three according to the realized performances and available processor resources, $\xi_d = i_{sd}^* - i_{sd}$ and $\xi_q = i_{sq}^* - i_{sq}$. $\max_n(\xi_d)$ and $\min_n(\xi_d)$ means the maximum and minimum value within $[\xi_d(k), \xi_d(k-1), \dots, \xi_d(k-n)]$, respectively, likewise on the q -axis component.

Discrete the model in (9) based on the Euler algorithm, the achieved model becomes the following:

$$\begin{aligned} \hat{e}(k+1) &= \alpha T_s u_s(k) + F_0 T_s + (F_1 T_s + 1) \hat{e}(k) \\ &\quad + T_s \hat{e}(k)^T F_2 \hat{e}(k) \\ &= F_u u_s(k) + F'_0 + F'_1 \hat{e}(k) + \hat{e}(k)^T F'_2 \hat{e}(k) \end{aligned} \quad (12)$$

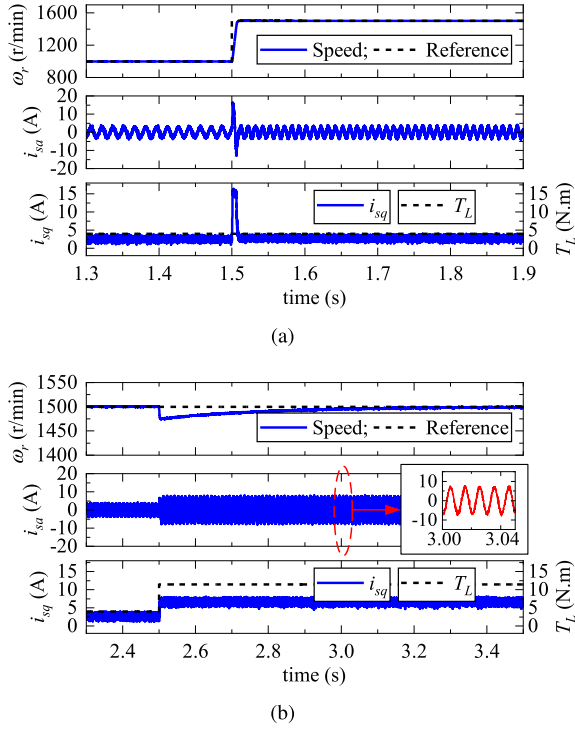


Fig. 3. Simulation waveforms for the proposed method. (a) Reference changing process. (b) Disturbance resisting process.

where $F_u = \alpha T_s$, $F'_0 = F_0 T_s$, $F'_1 = F_1 T_s + 1$, and $F'_2 = T_s F_2$.

The coefficient vector $\hat{\theta}$ and data vector ψ are defined as follows:

$$\begin{cases} \hat{\theta}_d(k) = [F_{ud}(k), F'_{0d}(k), F'_{1d}(k), F'_{2d}(k)]^T \\ \hat{\theta}_q(k) = [F_{uq}(k), F'_{0q}(k), F'_{1q}(k), F'_{2q}(k)]^T \end{cases} \quad (13)$$

$$\begin{cases} \varphi_d(k) = [u_{sd}(k), 1, \hat{e}_d(k), \hat{e}_d(k)^T \hat{e}_d(k)] \\ \varphi_q(k) = [u_{sq}(k), 1, \hat{e}_q(k), \hat{e}_q(k)^T \hat{e}_q(k)] \end{cases} \quad (14)$$

and are substituted into the RLS estimation algorithm. Considering the time shift of the estimation algorithm, the one-step-ahead prediction of the current tracking error is given by the following:

$$\hat{e}(k+1) = \psi_x(k+1)^T \hat{\theta}_x(k) \quad (15)$$

in which the subscript x means the component of d or q .

The reference is predicted to the $(k+1)$ th sampling period by the Lagrange interpolation theorem, and the estimated current is obtained by solving (11) and (15) as follows:

$$\begin{aligned} \hat{i}_s^*(k+1) &= i_s^*(k+1) + \min_n(\xi_x) \\ &\quad - [\max_n(\xi_x) - \min_n(\xi_x)] \hat{e}(k+1) \\ &\approx 3i_s^*(k) - 3i_s^*(k-1) + i_s^*(k-2) + \min_n(\xi_x) \\ &\quad - [\max_n(\xi_x) - \min_n(\xi_x)] \hat{e}(k+1). \end{aligned} \quad (16)$$

Since the proposed method belongs to the finite-control-set (FCS), the cost function is designed to be the same as (6) to realize the minimum error while tracking the current references.

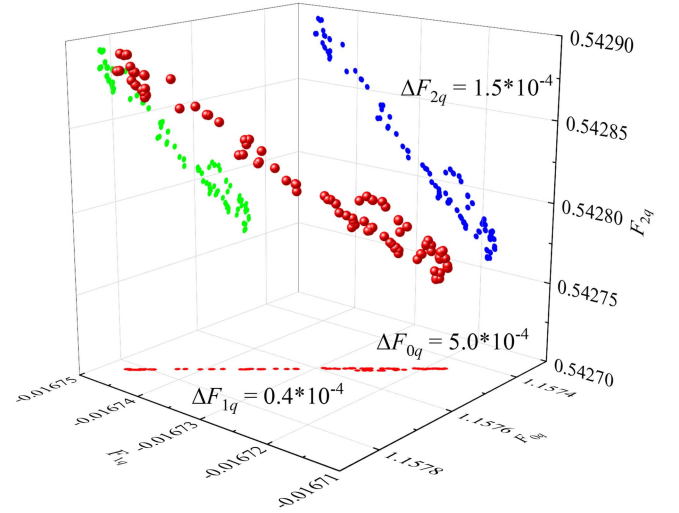


Fig. 4. Relative importance of affine operators in (9) for constant (F_0), linear (F_1), and square (F_2) components.

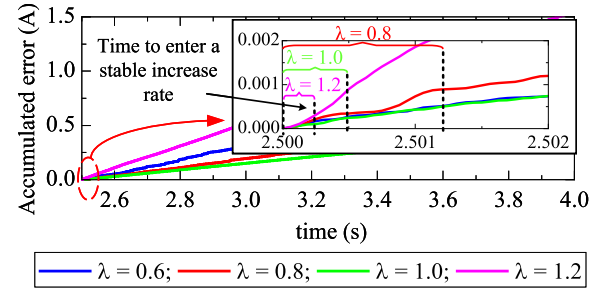


Fig. 5. Accumulated errors with different forgetting factors.

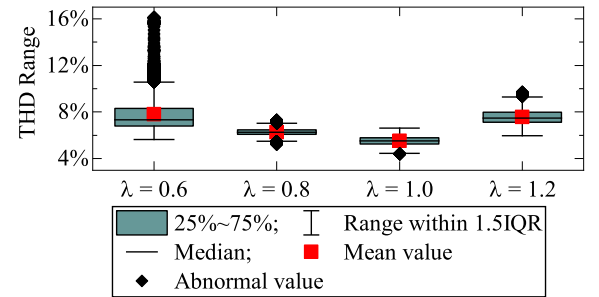


Fig. 6. Boxplot of THDs with different forgetting factors.

2) High-Order Extended Affine Ultralocal Structure: The abovementioned two-order structure suits the dq coordinate. However, the order requires to increase to three or more when the proposed method operates on the abc coordinate or other coordinates with higher dimensions. Considering possible tensors, the model in (9) could be improved as follows:

$$\frac{d\hat{e}}{dt} = \alpha u_s + F_0 + F_1 \hat{e} + \hat{e}^T F_2 \hat{e} + \hat{e}^T (F_3 \bar{x}_1 \hat{e}) \hat{e} + \dots \quad (17)$$

in which the error coupling terms in tensors $F_0, F_1, F_2, \dots$ are selected as zero, such as (10) to simplify the calculation

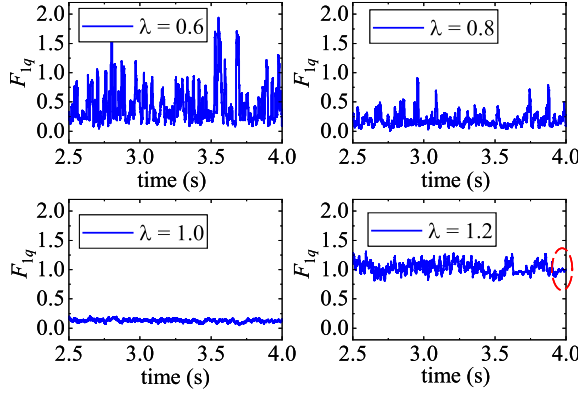


Fig. 7. Coefficient converging processes with different forgetting factors.

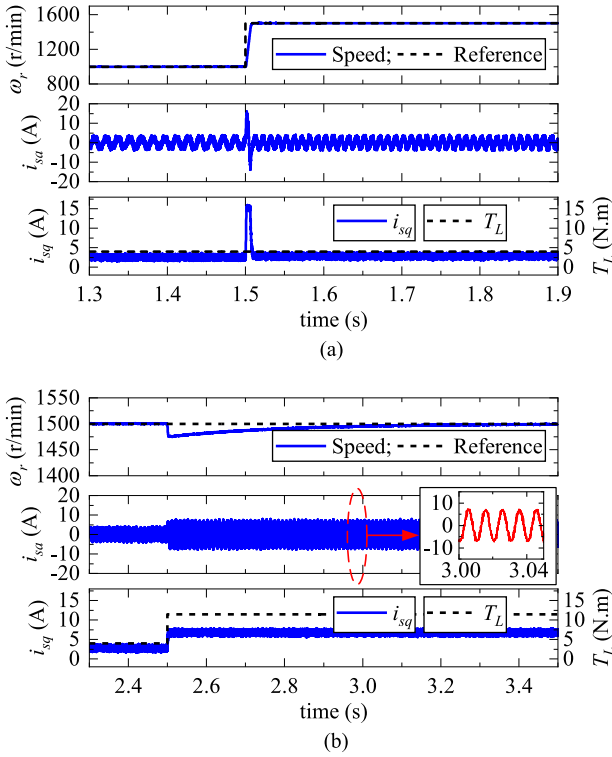


Fig. 8. Simulation waveforms for the proposed method on the abc coordinate. (a) Reference changing process. (b) Disturbance resisting process.

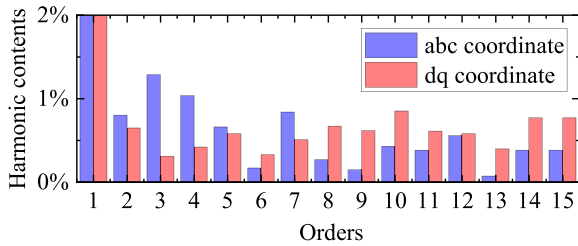


Fig. 9. Fourier results for the proposed method on the abc and dq coordinates.

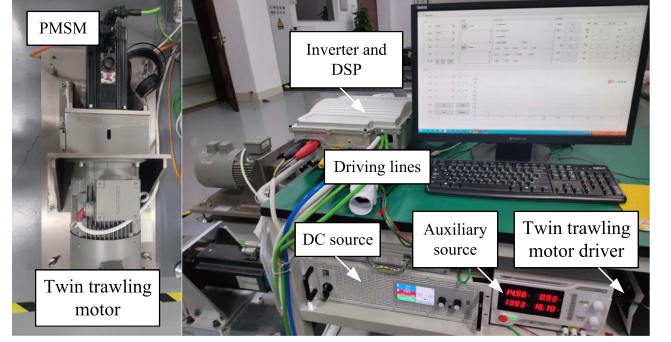


Fig. 10. Experimental platform.

TABLE III
PRIMARY SIMULATION PARAMETERS

Control performances	Prop. method	Conv. MF-PCC 1	Conv. MF-PCC 2	Conv. PCC
Settling time in disturbance-resisting process	6.4 s	6.7 s	6.5 s	6.8 s
Overshoot in disturbance-resisting process	18.5 %	19.0 %	18.5 %	19.0 %
Settling time in reference tracking process	1.7 s	1.9 s	1.5 s	2.0 s
Overshoot in reference tracking process	3.4 %	3.0 %	3.0 %	3.8 %
Phase current THD	3.143 %	4.558 %	4.929 %	3.883 %
Speed ITAE	0.1083	0.1781	0.1719	0.1287
Maximum ripple on q -axis current	1.587 A	1.977 A	1.785 A	1.429 A
Computation time	30.5 μ s	27.3 μ s	29.2 μ s	17.1 μ s

processes. These two-order or higher order sensitivity functions can effectively resolve the neglected influences on the affine operator in conventional affine arithmetic, caused by the nonlinear two-order and high-order terms [29].

The required parameters are given in Table I compared with the MF-PCC-based ARX with RLS algorithm in [22], MF-PCC-based ultralocal with LESO in [16], and conventional PCC strategy. The number of parameters is reduced to two and the dependence from all physical parameters of the motor is eliminated effectively for the proposed method.

IV. SIMULATION VALIDATIONS

The structure of the proposed method is shown in Fig. 2. Based on the structure, a simulation environment is built on MATLAB software, and the main parameters are given in Table II. The speed reference and the load torque are changed from 1000 to 1500 r/min at 1.5 s and from 4 to 11.5 N.m at 2.5 s, respectively.

A. Basic Performance and Forgetting Factor Selection

The simulation waveforms of speed, A-phase current, q -axis current, and load torque are shown in Fig. 3. It shows that the proposed method can track the reference and resist the

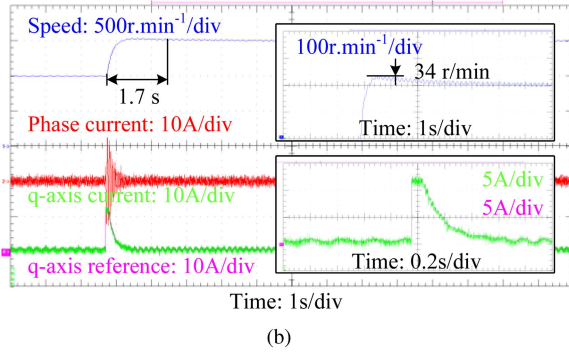
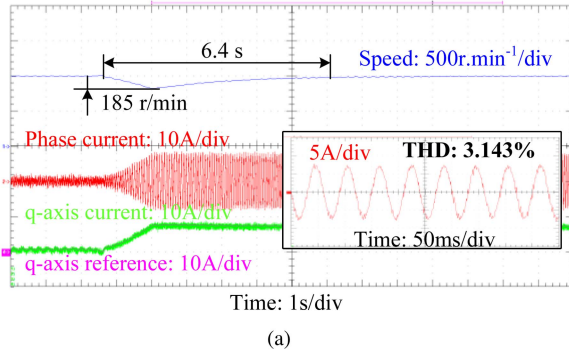


Fig. 11. Experimental waveforms for the proposed method. (a) Load torque disturbance rejection. (b) Reference changing process.

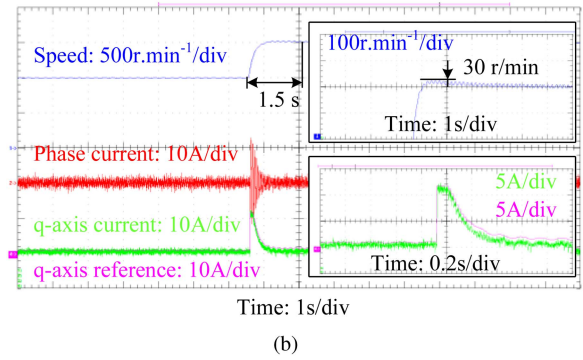
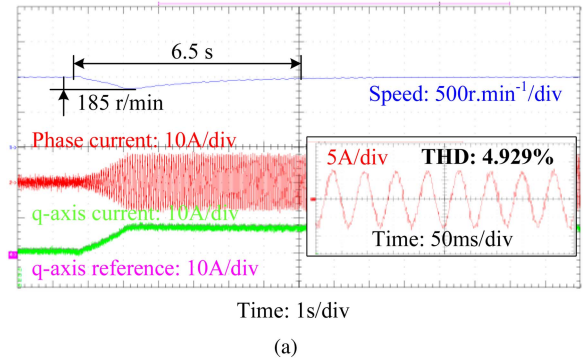


Fig. 13. Experimental waveforms for the conventional MF-PCC 2. (a) Load torque disturbance rejection. (b) Reference changing process.

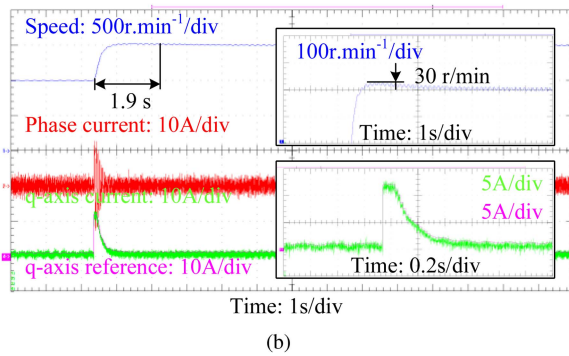
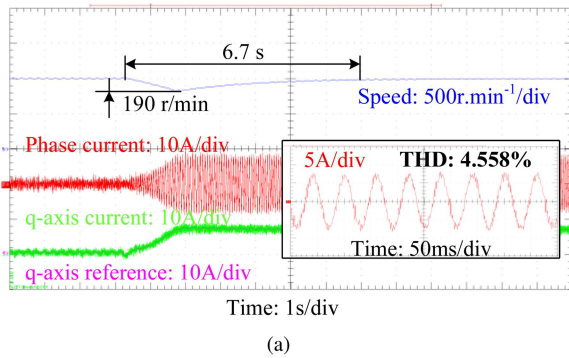


Fig. 12. Experimental waveforms for the conventional MF-PCC 1. (a) Load torque disturbance rejection. (b) Reference changing process.

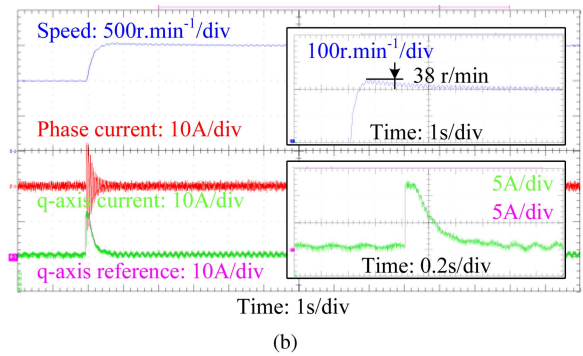
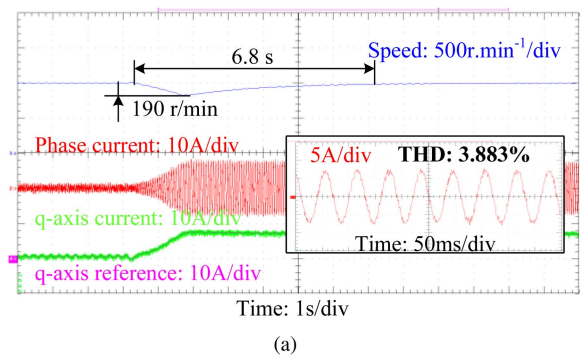


Fig. 14. Experimental waveforms for the conventional PCC strategy. (a) Load torque disturbance rejection. (b) Reference changing process.

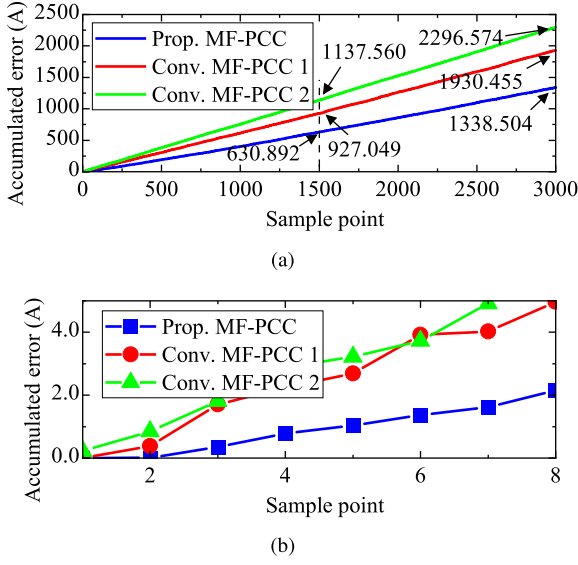


Fig. 15. Accumulated errors of the experimental currents on the dq coordinate for the proposed and conventional MF-PCCs. (a) Whole testing results. (b) Part enlarged testing results.

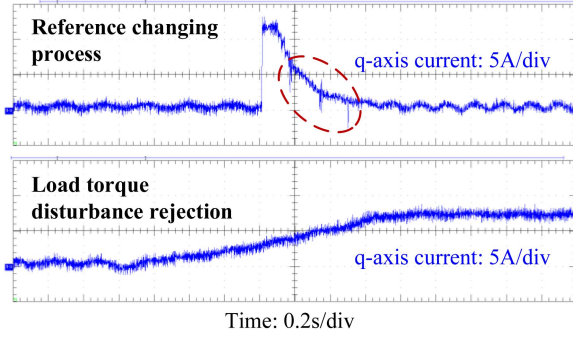


Fig. 16. Experimental waveforms of the q -axis current for the proposed method with the λ of 0.6.

disturbance successfully with a good sinusoidal degree of the phase current.

The affine operators F_0 , F_1 , and F_2 represent the significance of the constant term, affine variable ε , and the square of affine variable $\varepsilon^T \varepsilon$, respectively. The locations and rough ranges of these operators on the q -axis during the operating process are shown in Fig. 4 to verify their importance, in which the dots in red, green, and blue are the projection points of the locations. By considering the magnitudes of the locations, it is observed that the constant term is the most significant, followed by the square of the affine variable. This demonstrates the necessity of including nonlinear terms to improve model accuracy.

Since the forgetting factor λ in the RLS algorithm is a tradeoff between the high robustness against disturbances and fast tracking capability according to the expressions in [30], it is necessary to select its value according to a comprehensive performance analysis. The accumulated q -axis current errors with different forgetting factors are shown in Fig. 5. It shows that the minimum

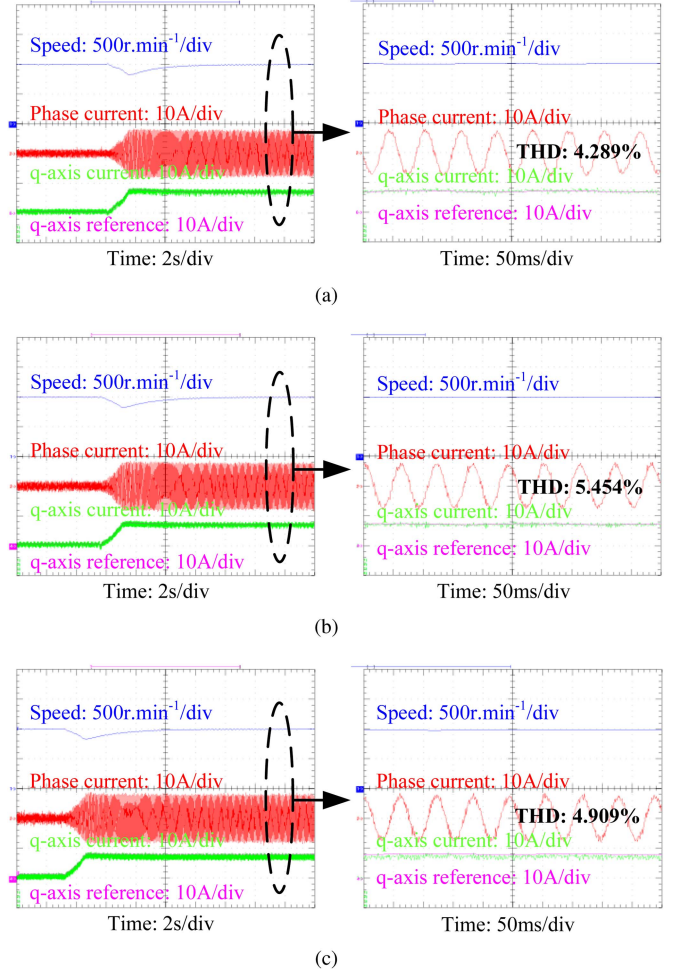


Fig. 17. Experimental waveforms under the conditions of parameter mismatches of $R_{se}/R_s = 0.1$, $L_{se}/L_s = 0.5$, and $\psi_{me}/\psi_m = 2$. (a) Proposed method. (b) Conventional MF-PCC 1. (c) Conventional MF-PCC 2.

current error is realized when the forgetting factor λ equals 1, and the best control accuracy is obtained. The errors will be increased with the increment or decrement of λ . According to the enlarged area of the graph, the time required for the error to reach a stable increase rate is longer for decreasing values of λ , i.e., the convergence rate is reduced. The convergence rate of $\lambda = 0.6$ is the lowest compared with the other waveforms because the error traces of $\lambda = 0.6$ and $\lambda = 1.0$ are almost overlapped at the beginning, and the slope of the former has been obviously exceeded after about 2.65 s.

The THDs within 3–4 s of the phase current are shown as a box diagram in Fig. 6. When $\lambda = 0.6$, a string of abnormal values of THD appears due to its low convergence rate, and these values basically disappear for the other λ values. According to the mean values of these THDs, they are increased after the initial drop with the increment of λ , and the best current quality is realized when $\lambda = 1$ with the lowest mean value.

The waveforms of coefficient F_{1q} with different λ values are shown in Fig. 7 with the same axis scales. It shows that the ripple amplitude is reduced by the decreasing $\lambda \leq 1$, and the

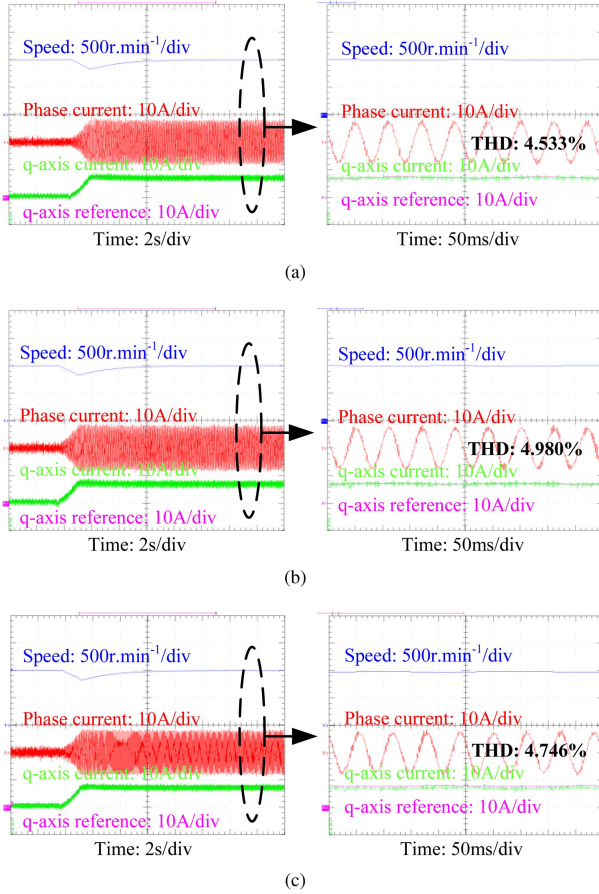


Fig. 18. Experimental waveforms under the conditions of parameter mismatches of $L_{se}/L_s = 2$ and $\psi_{me}/\psi_m = 2$. (a) Proposed method. (b) Conventional MF-PCC 1. (c) Conventional MF-PCC 2.

ripple is increased again by the increasing $\lambda > 1$. However, the convergence point circled in red has an obvious change, if λ is larger than one, and this change causes a large current error shown in Fig. 5 and low current quality in Fig. 6. The other coefficients in (13) have similar tendencies as F_{1q} for different λ values. Besides, the lengthy convergence processes within the RLS algorithm in [22] are effectively decreased by the proposed method, which further ensures stability.

B. Three-Order Extended Affine Ultralocal Structure in Three-Phase Stationary Coordinate

On the three-phase abc coordinate, the model order is increased as three, and the regressive matrices are all extended according to the model in (17). The simulation waveforms of speed, phase current, q -axis current, and load torque are shown in Fig. 8 under the same conditions, while the Fourier analyses of the phase currents are shown in Fig. 9 for the proposed method on both dq and abc coordinates. These figures indicate that on the abc coordinate, the proposed method is also capable of successfully tracking the reference and resisting disturbances. However, there is an increase in low-order harmonics, such as the second, third, and fourth orders, which may impact the

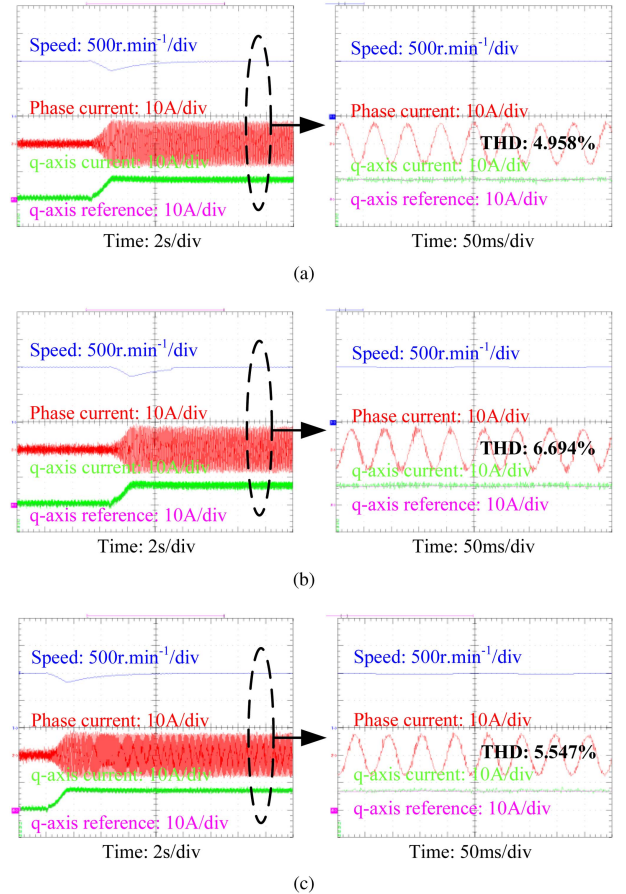


Fig. 19. Experimental waveforms under the conditions of parameter mismatches of $R_{se}/R_s = 10$, $L_{se}/L_s = 2$, and $\psi_{me}/\psi_m = 0.7$. (a) Proposed method. (b) Conventional MF-PCC 1. (c) Conventional MF-PCC 2.

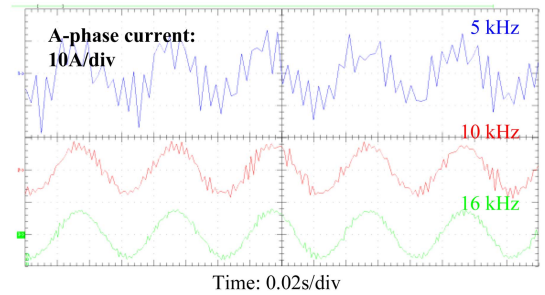


Fig. 20. Experimental waveforms of the A-phase current for the proposed method with different sampling periods.

sinusoidal degree of the stator current. To overcome this deficiency, further optimizing the coefficients in RLS or replacing this algorithm with a more accurate convergence estimation algorithm could be considered.

V. EXPERIMENTAL VALIDATIONS

The platform based on DSP F28379D and 4.8 kW PMSM is shown in Fig. 10, in which the parameters are the same as in Table II. A twin trawling motor is driven by an ABB driver of ACS880, which is connected to the shaft of the PMSM to

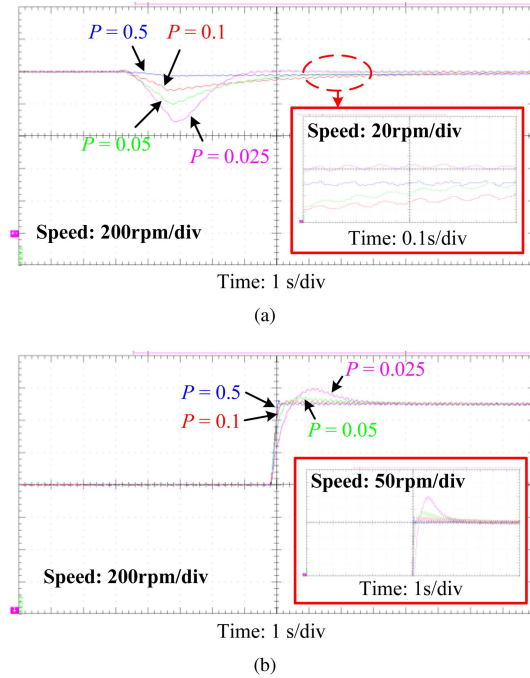


Fig. 21. Experimental waveforms of the speed for the proposed method with different proportional parameters in speed controller. (a) Load torque disturbance rejection. (b) Reference changing process.

provide a constant load torque. In addition, the sampled data is transmitted to the DSP for the purpose of performing estimation and modeling processes, and to generate control pulses, while the computer is used to connect to the serial communication interface terminal in the DSP for reading control variables and sending commands. In order to achieve better control performance, the sampling period is set to $62.5 \mu\text{s}$, which is a suitable value to prevent overrun errors occur from the processor.

A. Basic Control Performances

Under the conditions of no parameter mismatches, the experimental waveforms of speed, phase current, q -axis current, and its reference for the proposed method are shown in Fig. 11, in which the speed reference is changed from 1000 to 1500 r/min, and the load torque is changed from zero to 11.5 N.m. It shows that the system can reject the disturbance within 6.4 s, and the overshoot is about 18.5% during the disturbance rejection process. For the reference tracking process, the speed can stably track its reference within 1.7 s with an overshoot of about 3.4%. The phase current THD during the steady-state state with load is 3.143%.

Let the MF-PCC strategy based on the FCS-type conventional ultralocal with LESO in [16] and with ISMO in [31] be named as the conventional MF-PCC 1 and MF-PCC 2, respectively. The experimental waveforms of the conventional MF-PCC strategies and the conventional PCC strategy are shown in Figs. 12–14, respectively, under the same conditions. For the former, the settling times and overshoots are 6.7 s and 19.0% for the disturbance-resisting process, and 1.9 s and 3.0% for the reference tracking process. The phase current THD is 4.558%. The settling times

and overshoots are 6.5 s and 18.5% for the disturbance-resisting process, and 1.5 s and 3.0% for the reference tracking process, and the phase current THD is 4.929% for the middle. For the latter, the obtained overshoots and setting times are similar to the former strategy, and its phase current THD is 3.883%.

Some typical control performances are summarized and compared in Table III, in which the speed ITAE and maximum ripple on q -axis current are calculated within continuous 2000 sampled points during steady state. Since the speed controllers for the compared three strategies are all selected as the same one, they have similar dynamics for the compared three strategies, and the proposed method seems to improve response time, but the difference between the conventional methods and the proposed strategy is marginal. This could be basically explained by their similar converging rates without any noticeable error points in the comparison in Fig. 15, which shows the convergence processes of the proposed and conventional MF-PCCs during a transient state in experiments. Due to the improved model accuracy for the proposed method, it has the minimum phase current THD and speed ITAE, and they are reduced 31.04% and 39.19% compared with the conventional MF-PCC 1. Due to the good accuracy of the a priori model under the conditions of no parameter mismatches, the conventional PCC strategy gets the minimum q -axis ripple, lower THD, and right ITAE values. Moreover, the proposed method has the maximum computation time due to a higher number of values to be estimated in (10) and the matrix calculation in the RLS algorithm, and the orders of the extended affine ultralocal structure should be selected comprehensively considering the provided resources of the processor.

The experimental waveforms of the q -axis current for the proposed method with the λ of 0.6 are shown in Fig. 16. Compared with the other control strategies, some noises circled in red appear on the q -axis current due to the unsuitable value of λ in the RLS algorithm, and the converging process of the RLS algorithm might be affected.

B. Robustness Validations

Under the conditions of physical parameter mismatches, the conventional PCC strategy is used to generate initial training data of both MF-PCCs, and input gain in the conventional MF-PCC, to analyze the influence on the current quality of the MF-PCC strategies. Defining R_{se} , L_{se} , and ψ_{me} as the adopted values of the stator resistance, stator inductance, and magnet flux linkage in the controller, in which the ratio L_{se}/L_s operates on two components of the stator inductance at the same time. The parameter mismatches are set as $R_{se}/R_s = 0.1$, $L_{se}/L_s = 0.5$, and $\psi_{me}/\psi_m = 2$, and the experimental waveforms of the proposed method and the conventional MF-PCC strategies are shown in Fig. 17, including the combination of the enlarged q -axis currents. As shown in the figure, the dynamics of the proposed method and conventional MF-PCC strategy are similar to the waveforms without parameter mismatches. Compared with the conventional MF-PCC strategy, most ripple amplitudes on the q -axis current are reduced for the proposed method, and the realized THD of the phase current is also decreased.

Similar results appear in Figs. 18 and 19 of the experimental waveforms with parameter mismatches of $L_{se}/L_s = 2$ and $\psi_{me}/\psi_m = 2$, and $R_{se}/R_s = 10$, $L_{se}/L_s = 2$ and $\psi_{me}/\psi_m = 0.7$, respectively. The parameter mismatches have little effect on these three strategies because they do not require information about the values of time-varying parameters. The current qualities of the conventional strategies are slightly inferior to the proposed method due to the limited model accuracy of the ultralocal model.

C. Other Performance Validations

1) Sampling Period Selection: The experimental waveforms using sampling periods of 100 μs (sampling frequency of 10 kHz) and 200 μs (5 kHz) are shown in Fig. 20. From the figure, it can be observed that the quality of the current decreases significantly as the sampling period increases. This is attributed to the lack of accuracy in the model caused by insufficient data. When choosing the sampling period, it is important to consider both the required performance and the available processor resources. In the proposed method, a sampling period of 62.5 μs (sampling frequency of 16 kHz) is suggested as the minimum standard value, as it achieves good performance without any overrun errors.

2) Selection of Proportional Parameter in Speed Controller: In order to complete the dynamic validations and describe potential improvements for the proposed method, the experimental results with different proportional parameters, including 0.025, 0.05, 0.1, and 0.5, of the speed controller are shown in Fig. 21. As shown in the figure, the overshoots are obviously reduced when the proportional parameter is increased. However, the recovering process becomes longer, and settling times are increased, especially during the disturbance resisting process. Considering these indexes comprehensively, the selected parameter of 0.06 in the proposed method is suitable to obtain a balanced dynamics.

VI. CONCLUSION

To eliminate the influences of parameter mismatches and improve model accuracy, considering the influences of two-order term, an MF-PCC strategy based on extended affine ultralocal was presented in this article and applied in the PMSM driving system. A priori model of the motor in the model-based PCC was replaced by a data-driven model of extended affine ultralocal to reflect the motor motion characteristics, and the coefficients were online estimated by the RLS algorithm implying the input gain to accurately fit the current motor operating state. Compared with the conventional MF-PCC and conventional PCC strategies, the proposed method achieved good current quality and robustness under different conditions of parameter mismatches, and a proper forgetting factor was selected according to the simulation and experimental results.

To further improve the model accuracy, an error compensation term could be considered in future research, to upgrade the current and other objectives' qualities and be suitable for applications with stricter requirements.

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Yao Wei (Member, IEEE) was born in Handan, China, in 1993. He received the B.S. and Ph.D. degrees in power electronics and power transmission from Yanshan University, Qinhuangdao, China, in 2015 and 2021, respectively.

He is currently with Postdoc Project in material science and engineering, Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Jinjiang, Fujian, China. His research focuses on the driving and advanced control for electric vehicles

and servo systems.

Dr. Wei is currently a Senior Member of China Electrotechnical Society, and a Member of the Chinese Society for Electrical Engineering (CSEE).



Hector Young (Senior Member, IEEE) was born in Valparaíso, Chile, in 1984. He received the B.Eng. and M.Sc. degrees in electronics engineering from the Universidad de la Frontera, Temuco, Chile, both in 2009, and the Ph.D. degree in power electronics from the Universidad Técnica Federico Santa María, Valparaíso, Chile, in 2014.

Since 2014, he has been an Assistant Professor with Electrical Engineering Department, Universidad de La Frontera. His research inter-

ests include modeling and control of power converters and electrical drives, renewable energy systems, and microgrids.



Dongliang Ke (Senior Member, IEEE) was born in Quanzhou, China, in 1986. He received the B.S. degree in electrical engineering and the M.S. degree in power electronics and power drives from Northwestern Polytechnical University, Xi'an, China, in 2010 and 2013, respectively.

He is currently a Postdoctoral Supervisor, the Deputy Director of the Electrical Drives Laboratory, and a Senior Engineer with the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Jinjiang, Fujian, China, respectively. His research interests include predictive control, flux-oriented control, and sensorless control and identification for electrical drives and power converters.

Mr. Ke is an IET Member, and a Member of the Chinese Society for Electrical Engineering, and the Session Chair of the IEEE 5th International Symposium on Predictive Control of Electrical Drives and Power Electronics.



Fengxiang Wang (Senior Member, IEEE) was born in Jiujiang, China, in 1982. He received the B.S. degree in electronic engineering and the M.S. degree in automation from Nanchang Hangkong University, Nanchang, China, in 2005 and 2008, respectively, and the Ph.D. degree in electrical engineering from the Institute for Electrical Drive Systems and Power Electronics, Technische Universität München, Munich, Germany, in 2014.

He is currently a Full Professor and the Deputy Director of the Quanzhou Institute of Equipment Manufacturing, Haixi Institutes, Chinese Academy of Sciences, Jinjiang, Fujian, China. His research interests include predictive control and sensorless control for electrical drives and power electronics.

He was an IET Fellow, and an Associate Editor for IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS and IEEE TRANSACTIONS ON ENERGY CONVERSION. As the General Chair, he organized the IEEE 5th International Symposium on Predictive Control of Electrical Drives and Power Electronics.



José Rodríguez (Life Fellow, IEEE) received the engineer degree in electrical engineering from the Universidad Técnica Federico Santa María, Valparaíso, Chile, in 1977, and the Dr.-Ing. degree in electrical engineering from the University of Erlangen, Erlangen, Germany, in 1985.

Since 1977, he has been with the Department of Electronics Engineering, Universidad Técnica Federico Santa María, where he was a Full Professor and the President. From 2015 to 2019, he was the President of Universidad Andres Bello, Santiago, Chile. Since 2022, he has been the President of Universidad San Sebastián, Santiago, Chile. He has coauthored two books, several book chapters, and more than 700 journal and conference papers. His research interests include multilevel inverters, new converter topologies, control of power converters, and adjustable-speed drives.

Dr. Rodríguez was the recipient of a number of Best Paper Awards from journals of the IEEE, the National Award of Applied Sciences and Technology from the government of Chile, and in 2015, the Eugene Mittelmann Award from the Industrial Electronics Society of the IEEE. In 2021, he has been included in the list of Highly Cited Researchers published by Web of Science. He is a Member of the Chilean Academy of Engineering.